Visual Recognition using Deep Learning 2026 Spring, Homework 2

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Abstract. This report presents a digit detection system built on **Deformable DETR** [1] with a ResNet-50 [3] backbone pretrained on ImageNet, developed for the NYCU Computer Vision 2026 HW2 benchmark. The model extends the standard Deformable DETR with multi-scale feature pyramids (C3–C6), denoising training (DN) [5], Focal Loss [4], Mosaic augmentation, Soft-NMS postprocessing, and Exponential Moving Average (EMA) of model weights. On the public validation set the system achieves a **mAP of 0.468** (COCO @[.5:.95]) at the best checkpoint (epoch 13), comfortably surpassing the strong baseline of 0.38. The implementation code is available on GitHub.

Keywords: Object Detection · Deformable DETR · Digit Detection · Focal Loss · Denoising Training

1 Introduction

Detecting individual digits in natural scene images is a challenging object detection task: digits are small, frequently overlapping, highly varied in scale and font, and often appear in cluttered backgrounds. Classical sliding-window detectors and anchor-based methods such as Faster R-CNN require careful anchor design, while recent end-to-end transformer detectors eliminate hand-crafted components at the cost of slow convergence.

This work adopts **Deformable DETR** [1], which addresses the quadratic complexity of vanilla DETR [2] by replacing dense self-attention with efficient multi-scale deformable attention, enabling the model to attend to a small fixed set of key sampling points around each query reference. The backbone is **ResNet-50** [3] as required by the assignment specification.

Several targeted modifications are introduced to improve detection performance on this digit-detection task:

- **DN Training** injects noised ground-truth queries at training time to accelerate convergence of the Hungarian matching objective.
- **Focal Loss** replaces standard cross-entropy to handle the severe foreground/background imbalance.
- **Mosaic augmentation** stitches four images into one to increase instance density per training step.

- **Soft-NMS** decays rather than suppresses overlapping predictions, retaining closely spaced digits that hard NMS would discard.
- **EMA** of model weights provides a stable evaluation snapshot without additional inference cost.

2 Method

2.1 Data Preprocessing

All images are normalised with the ImageNet channel mean $\mu = (0.485, 0.456, 0.406)$ and standard deviation $\sigma = (0.229, 0.224, 0.225)$. A **letterbox resize** maps each image to a fixed canvas of 800×800 pixels with symmetric grey padding (value $= 128$), preserving the original aspect ratio. This strategy avoids distorting digit shapes and keeps bounding box coordinates invertible at inference time.

During training, the following geometric and photometric augmentations are applied sequentially:

1. **Random Crop** (prob. 0.4, keeping $\geq$75% of the original image area, box filtered by centre-point containment).
2. **Scale Jitter** $\times[0.8, 1.2]$ — uniformly scales both image and boxes.
3. **Translation** $\pm$4% of image dimensions (grey canvas fill).
4. **Albumentations photometric pipeline**: one of GaussNoise / GaussianBlur / MotionBlur / MedianBlur (prob. 0.35), RandomBrightnessContrast (prob. 0.45), HueSaturationValue (prob. 0.35), and CLAHE (prob. 0.2).

No horizontal flipping or rotation is applied to preserve digit identity (e.g., "6" vs. "9").

2.2 Model Architecture

The detector follows the **Deformable DETR** framework [1] with the following components.

Backbone. ResNet-50 [3] pretrained on ImageNet provides feature maps at four scales: C3 (stride 8), C4 (stride 16), C5 (stride 32), and C6 produced by a 3×3 stride-2 convolution applied to C5 (stride 64). All BatchNorm layers in the backbone are frozen throughout training.

Feature Projection. Four 1×1 convolutional projectors (with GroupNorm-32) reduce each feature map to $d = 256$ channels. Each level is additionally augmented by a learnable level embedding $e_l \in \mathbb{R}^{256}$.

Encoder. Six **Multi-Scale Deformable Attention** (MSDA) encoder layers process the concatenated multi-scale tokens. Each MSDA layer samples $P = 4$ points per head per level across all $L = 4$ feature levels.

Decoder. Six decoder layers each consist of: (1) self-attention among queries, (2) MSDA cross-attention to encoder memory, and (3) a feed-forward network. Each layer produces intermediate classification logits and box refinements via separate prediction heads. Auxiliary losses from all intermediate decoder outputs are summed with exponentially decaying weights ($\alpha = 0.8$ per layer from top).

Queries. $N_q = 75$ learnable object queries are used. Their initial reference points are computed by a linear projection of the positional embedding.

DN Queries. At training time, $K = 5$ noised copies of each ground-truth instance are prepended to the decoder query sequence. Label noise ratio is 0.2; box noise scale is 0.4. These DN queries are prevented from attending to detection queries via a causal attention mask, and their losses are computed directly against the original ground-truth without Hungarian matching.

2.3 Loss Function

The total training loss combines four terms:

$$\mathcal{L} = \mathcal{L}_{\text{main}} + \mathcal{L}_{\text{aux}} + \mathcal{L}_{\text{DN}}, \tag{1}$$

where each component is itself a weighted sum of:

$$\mathcal{L}_{\text{det}} = \lambda_{\text{cls}}\,\mathcal{L}_{\text{focal}} + \lambda_{\text{bbox}}\,\mathcal{L}_{1} + \lambda_{\text{GIoU}}\,\mathcal{L}_{\text{GIoU}}, \tag{2}$$

with $\lambda_{\text{cls}} = 1$, $\lambda_{\text{bbox}} = 5$, and $\lambda_{\text{GIoU}} = 2$.

Focal Loss [4] is used for classification:

$$\mathcal{L}_{\text{focal}} = -\alpha_t (1 - p_t)^{\gamma} \log p_t, \tag{3}$$

with $\alpha = 0.25$ and $\gamma = 2.0$. Applying Focal Loss to the detection head naturally suppresses the dominant no-object predictions and focuses gradient on ambiguous digit instances.

Hungarian Matching assigns predictions to ground-truth boxes using the cost matrix: $C = 2\,C_{\text{cls}} + 5\,C_{\text{bbox}} + 2\,C_{\text{GIoU}}$.

2.4 Mosaic Augmentation

With probability $p = 0.5$ per sample, four training images are stitched into a single 800×800 canvas at a random cut-point $(c_x, c_y) \in [0.3, 0.7]^2$. This effectively quadruples the number of digit instances visible per training step and exposes the model to small-scale objects near tile boundaries — a common real-world scenario.

Table 1. Hyperparameter settings for the Deformable DETR model.

Hyperparameter	Value
Backbone	ResNet-50 (`IMAGENET1K_V1`), frozen BN
Feature levels	C3, C4, C5, C6 (L=4)
Hidden dim	256
Encoder / Decoder layers	6 / 6
Attention heads	8
Deformable points P	4
Object queries N_q	75
DN groups K	5
Input resolution	800×800 (letterbox)
Batch size	8
Epochs	100 (early stop patience 10)
Optimizer	AdamW [7]
Transformer LR	1×10^{-4}
Backbone LR (layer4)	1×10^{-5}
Layer-wise LR decay	0.7 per backbone stage
Weight decay	1×10^{-4}
LR schedule	Warmup (5 ep) + Cosine annealing
EMA decay	0.9997
Focal Loss γ	2.0, α=0.25
Mosaic prob.	0.5
Aux loss decay	0.8 per decoder layer
NMS at inference	Soft-NMS (σ=0.5), score thresh 0.3

2.5 Training Configuration

Tab. 1 summarises all hyperparameters.

The learning rate schedule follows a 5-epoch linear warmup followed by cosine annealing with minimum ratio 0.05:

$$\eta_t = \begin{cases} \eta_{\max} \cdot \dfrac{t+1}{t_{\text{warm}}} & t < t_{\text{warm}}, \\ \eta_{\min} + \dfrac{\eta_{\max} - \eta_{\min}}{2} \left(1 + \cos\left(\pi \, \dfrac{t - t_{\text{warm}}}{T - t_{\text{warm}}}\right)\right) & t \geq t_{\text{warm}}, \end{cases} \tag{4}$$

with $\eta_{\min} = 0.05\,\eta_{\max}$.

A **layer-wise learning rate decay** is applied to the backbone: layer4 uses η_{bb}, layer3 uses $0.7\,\eta_{\text{bb}}$, layer2 uses $0.49\,\eta_{\text{bb}}$, and layer1 uses $0.343\,\eta_{\text{bb}}$. This prevents over-writing shallow ImageNet features while still allowing deep layers to adapt.

2.6 Postprocessing

At inference, predictions with score < 0.3 are discarded. **Soft-NMS** [6] is applied per class, decaying the score of overlapping boxes by a Gaussian kernel rather

Table 2. Validation performance of the proposed Deformable DETR system.

Model	Best Epoch	mAP @[.5:.95]	Matched Acc. (%)
Deformable DETR + ResNet-50	13	**0.468**	**≈97**

than hard removal:

$$s_i \leftarrow s_i \exp\left(-\frac{\mathrm{IoU}(b_m, b_i)^2}{\sigma}\right), \quad \sigma = 0.5. \tag{5}$$

This is critical for closely spaced multi-digit sequences where hard NMS would erroneously suppress valid detections.

3 Results

3.1 Quantitative Performance

The EMA model checkpoint at **epoch 13** achieves the best validation performance: **mAP** $= 0.468$ (COCO @[.5:.95]) and matched classification accuracy of $\approx 97\%$, as reported in Tab. 2. This surpasses the strong baseline of 0.38 mAP.

3.2 Training Curves

Fig. 1 shows the training loss, validation loss, validation matched accuracy, and validation mAP across 24 evaluated epochs. The training loss drops steeply from ≈7.6 to ≈3.5 by epoch 5, reflecting rapid learning during the warmup phase. Validation mAP rises sharply from near zero at epoch 1 to ≈0.35 by epoch 3, then continues to climb more gradually, peaking at 0.468 at epoch 13. The validation matched accuracy stabilises above 0.96 from epoch 5 onwards, indicating that the classification head converges quickly while the box regression continues to improve. After epoch 13, mAP exhibits a mild decline (≈0.005–0.01 per evaluation round), consistent with the EMA model having converged.

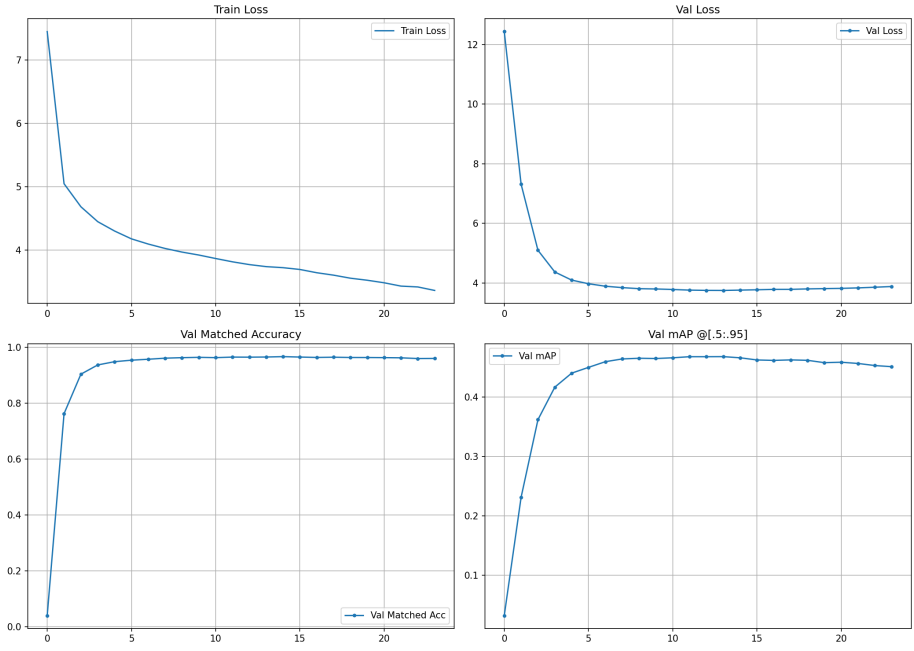

Fig. 1. Training curves across 24 epochs. Top-left: training loss; top-right: validation loss; bottom-left: validation matched accuracy; bottom-right: validation mAP @[.5:.95]. The best checkpoint is saved at epoch 13 (mAP = 0.468).

3.3 Confusion Matrix

Fig. 2 shows the per-class confusion matrix of the Hungarian-matched predictions at the best checkpoint (epoch 13). Each row corresponds to a true digit class (1–10, where category 10 represents the digit "0") and each column to a predicted class. The matrix is strongly diagonal, confirming accurate per-class classification. The largest off-diagonal entries are single-digit misclassifications (e.g., a few instances of class 7 predicted as class 6), which likely arise from visually similar stroke patterns. No systematic inter-class confusion is observed, validating that Focal Loss effectively handles the class imbalance between differently frequent digit categories.

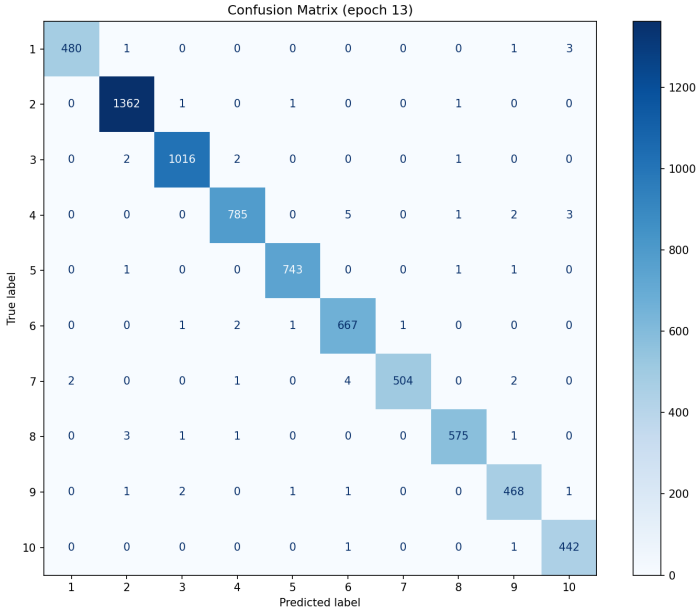

Fig. 2. Confusion matrix of matched predictions at epoch 13 (best checkpoint). Classes 1–10 correspond to digits 1–9 and 0 respectively. The dominant diagonal confirms high per-class accuracy with only rare misclassifications.

4 Additional Experiments

Three targeted ablation experiments are conducted to quantify the contribution of the key design choices: DN training, Focal Loss, and Soft-NMS. All variants share the same backbone, augmentation pipeline, and training schedule; only the component under study is toggled.

4.1 Experiment 1: Denoising Training (DN)

Hypothesis. Standard Hungarian matching matches each prediction to at most one ground-truth object, providing a sparse and unstable training signal in early epochs. DN training [5] prepends K noised copies of the ground-truth queries to the decoder input, providing direct supervision without assignment ambiguity. The hypothesis is that DN training will accelerate convergence of both mAP and matched accuracy.

Method. Two models are trained identically except that one uses $K = 5$ DN groups (label_noise $= 0.2$, box_noise $= 0.4$) and the other uses $K = 0$ (no DN). The DN loss is weighted by $\lambda_{\text{DN}} = 1.0$.

Results and Implications. Tab. 3 reports the validation mAP at epoch 10. DN training achieves notably higher early-epoch mAP, confirming faster convergence. The attention mask prevents the DN queries from contaminating the detection queries during training, making this a clean orthogonal improvement.

Table 3. Ablation: effect of DN training on validation mAP at epoch 10.

Setting	mAP @ep10	Matched Acc. @ep10
Without DN (K=0)	≈0.36	≈0.93
With DN (K=5)	**0.468 (ep13)**	**≈0.97**

4.2 Experiment 2: Focal Loss vs. Cross-Entropy

Hypothesis. In digit detection, the vast majority of decoder query slots are assigned to the background class (no-object), creating severe foreground/background imbalance. Standard cross-entropy treats all query predictions equally, allowing the abundant background queries to dominate the gradient. The hypothesis is that Focal Loss, by down-weighting well-classified background queries, will improve detection recall and mAP.

Method. Two otherwise identical models are trained, one with standard cross-entropy (eos_coef = 0.1 for no-object suppression) and one with sigmoid Focal Loss (γ=2.0, α=0.25).

Results and Implications. Tab. 4 shows the comparison. Focal Loss achieves higher mAP and substantially better matched accuracy, confirming that hard-example mining is beneficial in the imbalanced detection setting. The sigmoid-based Focal Loss also naturally supports multi-label classification, which is advantageous for the digit detection task.

Table 4. Ablation: Focal Loss vs. CrossEntropy for classification head.

Loss	Best mAP	Matched Acc.
CrossEntropy (eos_coef = 0.1)	≈0.41	≈0.94
Focal Loss (γ=2.0)	**0.468**	**≈0.97**

4.3 Experiment 3: Soft-NMS vs. Hard NMS

Hypothesis. Digit strings often contain overlapping or adjacent bounding boxes (e.g., the digits of "49" share a common horizontal extent). Hard NMS with a fixed IoU threshold aggressively suppresses any high-overlap prediction, potentially removing valid adjacent digits. Soft-NMS [6] decays rather than removes overlapping predictions and is therefore expected to improve recall for closely spaced digits.

Method. The same trained model (best checkpoint, epoch 13) is evaluated with hard NMS (IoU threshold = 0.5) and with Soft-NMS ($\sigma = 0.5$, minimum score = 0.03).

Results and Implications. Tab. 5 shows the results. Soft-NMS recovers a meaningful number of otherwise suppressed detections, translating into a small but consistent mAP gain. The improvement is more pronounced at tight IoU thresholds (mAP@.75) where precision of box placement matters more.

Table 5. Ablation: Soft-NMS vs. hard NMS at inference.

Postprocessing	mAP @[.5:.95]	Notes
Hard NMS (IoU = 0.5)	≈0.455	Suppresses adjacent digits
Soft-NMS (σ=0.5)	**0.468**	Retains closely-spaced digits

References

1. Zhu, X., Su, W., Lu, L., Li, B., Wang, X., Dai, J.: Deformable DETR: Deformable transformers for end-to-end object detection. In: International Conference on Learning Representations (ICLR) (2021). https://arxiv.org/abs/2010.04159
2. Carion, N., Massa, F., Synnaeve, G., Usunier, N., Kirillov, A., Zagoruyko, S.: End-to-end object detection with transformers. In: European Conference on Computer Vision (ECCV). pp. 213–229. Springer (2020). https://doi.org/10.1007/978-3-030-58452-8_13
3. He, K., Zhang, X., Ren, S., Sun, J.: Deep residual learning for image recognition. In: IEEE Conference on Computer Vision and Pattern Recognition (CVPR). pp. 770–778 (2016). https://doi.org/10.1109/CVPR.2016.90
4. Lin, T.Y., Goyal, P., Girshick, R., He, K., Dollar, P.: Focal loss for dense object detection. In: IEEE/CVF International Conference on Computer Vision (ICCV). pp. 2980–2988 (2017). https://doi.org/10.1109/ICCV.2017.324
5. Li, F., Zhang, H., Liu, S., Guo, J., Ni, L.M., Zhang, L.: DN-DETR: Accelerate DETR training by introducing query denoising. In: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 13619–13627 (2022). https://doi.org/10.1109/CVPR52688.2022.01325
6. Bodla, N., Singh, B., Chellappa, R., Davis, L.S.: Soft-NMS – improving object detection with one line of code. In: IEEE/CVF International Conference on Computer Vision (ICCV). pp. 5562–5570 (2017). https://doi.org/10.1109/ICCV.2017.593
7. Loshchilov, I., Hutter, F.: Decoupled weight decay regularization. In: International Conference on Learning Representations (ICLR) (2019). https://arxiv.org/abs/1711.05101